Quiz 2

September 10, 2026

1. In the following, let $T_s^*(n)$ denote the optimal running time of a problem (the problem itself is left implicit in discussions).

   (a) What are the important characteristics of good parallel solution on the WS model? Explain in details why they are important.

   (b) Between two WS algorithms, one of which has work $O(T_s^*(n))$ and span $O(\sqrt{n})$, while another has work $O(T_s^*(n) \cdot logn)$ and span $O(\log n)$. Which WS algorithm is considered the better algorithm? Give arguments to support your answer.

   (c) Between two WS algorithms, one of which has work $O(T_s^*(n) \cdot \log n)$ and span $\log n$, while another has work $O(T_s^*(n))$ and span $2 \log n$. Which WS algorithm is considered the better algorithm? Give arguments to support your answer.

   (d) Can we design a WS algorithm with work $o(T_s^*(n))$? Give reasons to support your answer.

2. We are to devise a WS algorithm for finding a maximum element of n numbers.

   (a) Given that our designed algorithm is optimal, what would be the work complexity and step complexity of the algorithm?

   (b) Describe WS code of the optimal WS algorithm.

3. Given n integers as input, you task here is to design a sequential algorithm as well as a parallel algorithm on the PRAM that multiply all the given integers as fast as you can. You are now to answer the following questions:

   (a) Describe an optimal sequential algorithm and express the resources used by the algorithm in terms of the running time of the algorithm. Also, argue that why the algorithm is optimal.

   (b) Describe an optimal parallel algorithm and express the resources used by the algorithm in terms of the running time and the number of processors. Also, argue that why the algorithm is optimal.